

Name:

Points: /20

Databasutveckling och lagring — Final Exam: RETAKE II

2025-03-06

Instructions

1. **No computers, no phones, no notes**
2. **No talking with other people in the room**
3. In case point 1 or 2 are broken, you get automatically **0 points** on the test.
4. Write your name on top of the page (do not fill in the *Points* box!)
5. For each question, you'll need to answer the question by crossing with an **X** the right answer (multiple choice questions) or write your answer in the empty lines below the question (open questions).
6. Once you're done:
 - Double-check your answers
 - Hand over your test to me
 - Leave the room

NB: While inside the classroom, rule 1 and 2 are still valid, even if you've already handed over your answers to me. If you want to discuss your answers with a classmate, or to check an answer on your phone, do that **outside the classroom**.

NB: Make sure to use the restroom, fill our water bottle, grab a coffee, etc... **before** we begin the exam, as you're not allowed to leave the classroom without handing over your answers.

Point system

There are **10 questions**, worth **2 points** each, for a total of maximum **20 points**.

There are no partial points

You either give the **complete, correct answer**, and get 2 points, or get 0 points.

Grading

Points	Grade
0 - 12	IG
13 - 19	G
20	VG

Name: [

READ EVERYTHING CAREFULLY!

1 Normalization

Given the following schema of a SQLite database:

Listing 1: The complete schema for the DB

```
PRAGMA foreign_keys = ON;
```

```
CREATE TABLE IF NOT EXISTS customer (  
    customer_id INTEGER PRIMARY KEY,  
    full_name VARCHAR(100) NOT NULL,  
    email VARCHAR(100) UNIQUE NOT NULL,  
    postal_code VARCHAR(5) NOT NULL,  
    city VARCHAR(50) NOT NULL  
);
```

```
CREATE TABLE IF NOT EXISTS orders (  
    order_id INTEGER PRIMARY KEY,  
    order_date TEXT NOT NULL,  
    customer_id INTEGER NOT NULL,  
    FOREIGN KEY (customer_id) REFERENCES customer(customer_id)  
);
```

```
CREATE TABLE IF NOT EXISTS products (  
    product_id INTEGER PRIMARY KEY,  
    product_name TEXT NOT NULL,  
    category TEXT,  
    unit_price FLOAT NOT NULL  
);
```

```
CREATE TABLE IF NOT EXISTS order_items (  
    order_id INTEGER NOT NULL,  
    product_id INTEGER NOT NULL,  
    quantity INTEGER NOT NULL,  
    product_name TEXT NOT NULL,  
    PRIMARY KEY (order_id, product_id),  
    FOREIGN KEY (order_id) REFERENCES orders(order_id),  
    FOREIGN KEY (product_id) REFERENCES products(product_id)  
);
```

Point out and explain **one** normalization problem present in the schema.

One normalization problem present in the schema is that the order_items table has a composite primary key (order_id, product_id) which is not a foreign key reference to the orders table. This is a violation of the first normal form (1NF) because the primary key is not a foreign key reference to the orders table.



- 2 Explain how you would solve the normalization problem you described above.

You don't need to write a SQL query, just explain how you would fix it.

By Avoid Insert , update , delete

- 3 In this course we've labbed with a database that contained the data for a music store. What was the name of the database, and which RDBMS did it use?

Name: _____

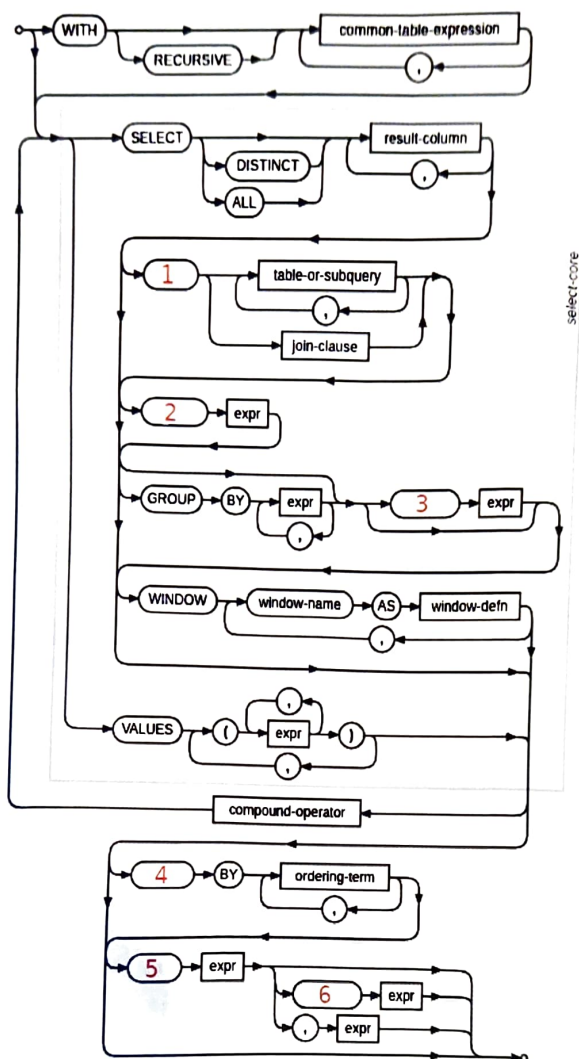
RDBMS: SQLite

- 4 In this course, we've labbed with a database that contained the data for a movie rental store. What was the name of the database, and which RDBMS did it use?

Name: _____

RDBMS: ManiaDB

5 Fill the blanks



1: Select

2: values with

3: Values

4: Group

5: Values

6: _____

6 What's the difference between WHERE and HAVING in a SELECT query?



7 What's the difference between 'JOIN...USING' and 'JOIN...ON'?



8 We can always replace 'JOIN...ON' with 'JOIN...USING'

- 
1. True
 2. False 

9 We can always replace 'JOIN...USING' with 'JOIN...ON'

- 
1. True 
 2. False



10 What is a subquery?

X

